

The Ethical Hacker's Command Handbook

Chapter 2 – Part 5: OS Detection & Advanced Detection Techniques

Learn how Nmap estimates operating systems using TCP/IP fingerprinting.

Learning Objectives

Understand OS detection, fingerprinting, OS scan options, accuracy, limitations, workflows, and best practices.

What is OS Detection?

OS detection estimates the target operating system by comparing network responses to Nmap's fingerprint database. It helps identify platforms such as Windows, Linux, BSD, or network appliances.

Why It Matters

Knowing the operating system helps guide enumeration, service analysis, and vulnerability research during authorized security assessments.

sudo nmap -O

Purpose: Detect operating system

What it does: Compares probe responses with known fingerprints.

When to use: After discovering open ports.

Avoid when: Only checking host availability.

Memory Tip: O = Operating System.

sudo nmap --osscan-limit

Purpose: Limit OS detection

What it does: Attempts OS detection only on suitable hosts.

When to use: Large network scans.

Avoid when: Small single-host scans where speed is not a concern.

Memory Tip: Reduces unnecessary probes.

sudo nmap --osscan-guess

Purpose: Best OS guess

What it does: Shows closest fingerprint match when confidence is low.

When to use: Lab environments or uncertain results.

Avoid when: When exact identification is required.

Memory Tip: Guess = Best estimate.

sudo nmap -O -sV

Purpose: OS + Version detection

What it does: Combines OS fingerprinting with service version detection.

When to use: Most detailed manual enumeration.

Avoid when: Quick host discovery.

Memory Tip: Versions support OS analysis.

sudo nmap -A

Purpose: Aggressive scan

What it does: Runs OS detection, version detection, default NSE scripts, and traceroute.

When to use: Small lab machines after initial enumeration.

Avoid when: Large networks or quick reconnaissance.

Memory Tip: Use after basic scans.

Why OS Detection Can Fail

Common reasons include firewall filtering, packet loss, VPNs, custom TCP/IP stacks, and the absence of both open and closed ports. Treat results as informed estimates rather than absolute proof.

Comparison Table

Goal	Command
Detect OS	-O
Limit OS detection	--osscan-limit
Best estimate	--osscan-guess
OS + Versions	-O -sV
Comprehensive scan	-A

Typical Workflow

Target → -sS → -sV → -O → Research → Targeted Enumeration → Document Findings

Revision Notes

- -O = Detect OS
- --osscan-limit = Suitable hosts only
- --osscan-guess = Closest match
- -O -sV = OS + Versions
- -A = OS + Versions + Default Scripts + Traceroute

Practice Questions

1. What does -O do?
2. Why can OS detection fail?
3. When should you use --osscan-limit?
4. What is the purpose of --osscan-guess?
5. Why combine -O with -sV?